

1. A 24-year-old woman receives a 6-day course of oral ampicillin for acute cystitis. One week later, she has vaginal pruritus, burning, and dyspareunia. Which of the following organisms is the most likely cause of these symptoms?

- ☐ A) *Candida albicans*
- ☐ B) *Gardnerella vaginalis*
- ☐ C) *Mycoplasma hominis*
- ☐ D) *Neisseria gonorrhoeae*
- ☐ E) *Trichomonas vaginalis*



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Lab Values



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Review



Help



Pause

2. A 27-year-old primigravid white woman at 9 weeks' gestation comes to the physician for her first prenatal visit. There is no family history of genetic disorders, thrombophilia, or intellectual developmental disorder. Examination shows a uterus consistent in size with a 9-week gestation. Which of the following is the most appropriate recommendation for this patient?
- ☐ A) Cystic fibrosis screening
 - ☐ B) Diabetes mellitus screening
 - ☐ C) Serum lipid studies
 - ☐ D) Thalassemia screening
 - ☐ E) Chorionic villus sampling

3. A 24-year-old primigravid woman at 25 weeks' gestation is admitted to the hospital because of a 24-hour history of absent fetal movement. Pregnancy had been uncomplicated with no contractions, vaginal spotting, or loss of vaginal fluid. She has received routine prenatal care. Vital signs are within normal limits. Physical examination shows no abnormalities. Fundal height is 25 cm. Fetal heart tones cannot be heard on auscultation. Which of the following is most likely to confirm the diagnosis?
- ☐ A) Fetal nonstress test
 - ☐ B) Kleihauer-Betke (acid elution) test
 - ☐ C) Measurement of maternal serum β -hCG concentration
 - ☐ D) Fetal ultrasonography
 - ☐ E) Amniocentesis

4. A 42-year-old nulligravid woman comes to the physician because she has been unable to conceive for 16 months. She has had unprotected sexual intercourse with her partner four times weekly during this time. Her last menstrual period was 10 days ago. Menses occur at regular 28- to 30-day intervals and last 4 days. She has not had nipple discharge or changes in her weight. She has no history of serious illness, operative procedures, or sexually transmitted diseases. She takes no medications. Physical examination, including pelvic examination, shows no abnormalities. Hysterosalpingography and semen analysis of her partner show no abnormalities. Results of serum studies obtained on day 3 of the patient's menstrual cycle are most likely to show an increase in which of the following concentrations?
- ☐ A) Estradiol
 - ☐ B) Follicle-stimulating hormone
 - ☐ C) Luteinizing hormone
 - ☐ D) Prolactin
 - ☐ E) Thyroid-stimulating hormone

5. A 31-year-old woman, gravida 3, para 3, comes to the physician for a postpartum examination 6 weeks after cesarean delivery at 26 weeks' gestation. Her pregnancy was complicated by preterm labor. The infant is improving in the intensive care unit. Since delivery, the patient has had fatigue, insomnia, difficulty eating, and frequent episodes of crying. She has been attempting to use a breast pump but has had poor milk production. Her husband is in the military and is currently deployed overseas for 6 months. Her two other children are from a previous relationship and live with their father. Physical examination shows a well-healed incision. On mental status examination, she has a flat affect and is intermittently tearful. She says she feels the nursery staff do not like her. She is afraid that her husband will not return from deployment. Her Edinburgh Postnatal Depression Scale score is 20/30 (Normal<10). Which of the following is the most likely diagnosis?
- ☐ A) Bipolar disorder
 - ☐ B) Generalized anxiety disorder
 - ☐ C) Major depressive disorder
 - ☐ D) Postpartum psychosis
 - ☐ E) Schizoaffective disorder
 - ☐ F) Normal grief reaction

6. A 15-year-old girl comes to the emergency department because of pain in the midline and right abdomen for 3 hours. Her last menstrual period was 7 weeks ago. She has had nausea and vomiting for 1 hour. Her temperature is 37.8°C (100°F), pulse is 90/min, and blood pressure is 130/68 mm Hg. Abdominal examination shows severe tenderness in the right lower quadrant. Ultrasonography shows an ill-defined 4-cm mass in the right lower quadrant. A pregnancy test is positive. Which of the following is the most appropriate next step in management?
- ☐ A) Admission to the hospital for laparoscopy
 - ☐ B) Admission to the hospital for laparotomy
 - ☐ C) CT scan of the pelvis
 - ☐ D) Limited-stay observation
 - ☐ E) MRI of the pelvis
 - ☐ F) Uterine evacuation in the operating room

7. An otherwise healthy 42-year-old woman comes to the physician because of a 1-year history of irregular menses, hot flashes, occasional night sweats, and irritability. Her last menstrual period was 5 months ago. Menses had previously occurred at irregular 8- to 13-week intervals. She is sexually active and uses condoms for contraception. She has had no other symptoms and takes no medications. She has two children, ages 14 and 18 years. She is 163 cm (5 ft 4 in) tall and weighs 66 kg (146 lb); BMI is 25 kg/m². Physical examination shows no abnormalities. Pelvic examination shows a slightly enlarged uterus and normal ovaries. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Measurement of serum estradiol concentration
 - ☐ B) Measurement of serum follicle-stimulating hormone concentration
 - ☐ C) Measurement of serum β -hCG concentration
 - ☐ D) Measurement of serum thyroid-stimulating hormone concentration
 - ☐ E) Progestin challenge test



8. A 15-year-old girl is brought to the physician because of heavy vaginal bleeding and passage of plum-sized clots over the past 7 days. Her last menstrual period was 2½ months ago. She is not sexually active. She is 168 cm (5 ft 6 in) tall and weighs 91 kg (200 lb); BMI is 32 kg/m². Breast and pubic hair development is sexual maturity rating stage 4. Examination shows a 50-mL clot in the vaginal vault; the cervical os is closed. The uterus is normal in size and contour. Her hemoglobin concentration is 10 g/dL. Which of the following is the most appropriate treatment for this patient?
- ☐ A) Clomiphene therapy
 - ☐ B) Leuprolide therapy
 - ☐ C) Oral contraceptive therapy
 - ☐ D) Dilatation and curettage
 - ☐ E) No treatment indicated

9. A 27-year-old woman, gravida 2, para 1, at 40 weeks' gestation is admitted to the hospital in labor. Contractions are moderate and occur every 3 minutes. She has received routine prenatal care. She has no history of serious illness. She has smoked one pack of cigarettes daily for 10 years and has continued to smoke throughout pregnancy. A fetal heart tracing is shown. The cervix is 100% effaced and 5 cm dilated; the vertex is at 0 station. The membranes are intact. Which of the following is the most likely cause of this fetal heart pattern?

- ☐ A) Cervical dilation
- ☐ B) Fetal head compression
- ☐ C) Fetal movement
- ☐ D) Umbilical cord compression
- ☐ E) Uteroplacental insufficiency



10. A 32-year-old woman comes to the physician 1 week after she noticed a tender lump in her right breast during routine breast self-examination. She has no history of serious illness and takes no medications. Menses occur at regular 28-day intervals, and her last menstrual period was 2 weeks ago. Examination shows a 3-cm, tender mass in the upper outer quadrant of the right breast; there is no nipple discharge. Ultrasonography of the right breast shows a cystic lesion. Which of the following is the most appropriate next step in management?
- ☐ A) Second breast examination in 6 months
 - ☐ B) Mammography
 - ☐ C) Fine-needle aspiration of the mass
 - ☐ D) True-cut needle biopsy
 - ☐ E) Lumpectomy

11. A 42-year-old woman at 28 weeks' gestation is admitted to the hospital because of severe preeclampsia. Genetic testing at 16 weeks' gestation showed a fetus with trisomy 18; she refused pregnancy termination at that time. Her pulse is 84/min, respirations are 16/min, and blood pressure is 160/110 mm Hg. Examination today shows a uterus consistent in size with a 28-week gestation, a nontender abdomen, a long and closed cervix, mild dependent edema, and normal deep tendon reflexes. The physician recommends induction of labor, and the patient's husband urges her to follow the physician's recommendation. She refuses because under no circumstances does she want to give birth to a premature infant. One day later, she has a generalized tonic-clonic seizure. She remains alert and continues to refuse labor induction. Which of the following is the most appropriate course of action?
- ☐ A) Await spontaneous labor because of the patient's refusal of labor induction
 - ☐ B) Await spontaneous labor unless the patient's situation worsens
 - ☐ C) Deliver the fetus because her husband will provide consent on her behalf
 - ☐ D) Deliver the fetus because the patient's life is threatened
 - ☐ E) Deliver the fetus because trisomy 18 is a terminal condition

12. A 37-year-old woman, gravida 2, para 1, at 10 weeks' gestation comes to the physician for her first prenatal visit. Seven years ago, she underwent cone biopsy of the cervix for cervical cancer. Four years ago, her first pregnancy ended in preterm delivery at 20 weeks' gestation following a 3-hour history of strong, painless pelvic pressure and vaginal bleeding; the newborn died shortly after delivery. Pelvic examination today shows no abnormalities. Ultrasonography shows a fetus consistent in size with a 10-week gestation, a cervical length of 4 cm, and a low anterior placenta. This patient is at greatest risk for which of the following conditions?
- ☐ A) Abruptio placentae
 - ☐ B) Cervical cancer
 - ☐ C) Cervical insufficiency
 - ☐ D) Cervical stenosis
 - ☐ E) Placenta previa

13. A 27-year-old nulligravid woman comes to the physician for a follow-up examination after a recent Pap smear showed a low-grade squamous intraepithelial lesion. She feels well and has no history of sexually transmitted diseases or serious illness. Menses occur at regular 30-day intervals. Her last menstrual period ended 4 weeks ago. Physical and pelvic examinations show no abnormalities. Colposcopic examination shows punctation and a mosaic pattern; the squamocolumnar junction cannot be visualized. A biopsy specimen of the cervix shows cervical intraepithelial neoplasia (CIN) 3. Which of the following is the most appropriate next step in management?
- ☐ A) Pap smear every 6 months
 - ☐ B) Hysteroscopy
 - ☐ C) Cone biopsy of the cervix
 - ☐ D) Endometrial biopsy
 - ☐ E) Intravaginal radiation therapy
 - ☐ F) Hysterectomy

14. A previously healthy 32-year-old woman comes to the physician because of a heavy menstrual period 1 week ago. She says that her menstrual period was 1 week late and she had vaginal spotting 2 days prior to its onset. Menses typically occur at regular 28- to 29-day intervals and last 4 days. She is sexually active, and she and her partner use spermicidal foam and condoms for contraception. She takes no medications. Pelvic examination shows a normal uterus. There are no adnexal masses. Which of the following is the most appropriate next step in management?
- ☐ A) Measurement of serum follicle-stimulating hormone concentration
 - ☐ B) Measurement of serum β -hCG concentration
 - ☐ C) Progesterone suppositories
 - ☐ D) Hysteroscopy
 - ☐ E) Dilatation and curettage

15. A 57-year-old woman, gravida 2, para 2, comes to the physician because of a 6-month history of worsening incontinence. Ten years ago, she had leaking of urine only with laughing or jogging. Now, she also has urine leakage when rising from a sitting position. She says this is interfering with her usual activities, and she has to wear absorbent pads all the time. She takes amitriptyline for migraine prophylaxis. Vital signs are within normal limits. Pelvic examination shows a normal-appearing cervix but decreased rugation of the vaginal mucosa. The urethra appears normal. On Valsalva maneuver, the anterior vaginal wall bulges inferiorly. Which of the following is the most likely cause of this patient's incontinence?
- ☐ A) Adverse effect of amitriptyline
 - ☐ B) Detrusor overactivity
 - ☐ C) Estrogen deficiency
 - ☐ D) Impaired detrusor compliance
 - ☐ E) Inadequate urethral support
 - ☐ F) Intrinsic sphincter deficiency

16. A healthy 19-year-old woman comes to the physician for advice regarding contraception. She plans on becoming sexually active for the first time with her partner soon. Menarche was at the age of 13 years. Menses occur at regular 28-day intervals. She does not smoke cigarettes. She is 165 cm (5 ft 5 in) tall and weighs 54 kg (120 lb); BMI is 20 kg/m². Her blood pressure is 110/68 mm Hg. Physical examination shows no abnormalities. She declines a speculum or pelvic examination. In addition to providing the patient with condoms, which of the following is the most appropriate next step in management?
- ☐ A) Provide the patient with information on pelvic examinations
 - ☐ B) Recommend the patient return in 6 months
 - ☐ C) Schedule an examination with a pediatric gynecologist
 - ☐ D) Do not prescribe contraception until pelvic examination
 - ☐ E) Prescribe an oral contraceptive for 3 months and schedule a follow-up examination

17. A previously healthy 24-year-old woman, gravida 1, para 1, comes to the physician 2 hours after the sudden onset of severe pain, numbness, and stiffness in her right leg. Three weeks ago, she delivered a healthy newborn at term after an uncomplicated pregnancy. Her temperature is 37°C (98.6°F), pulse is 70/min and regular, respirations are 14/min, and blood pressure is 110/70 mm Hg. The lungs are clear to auscultation. S₁ is normal. S₂ is split and fixed. A grade 2/6 holosystolic murmur is heard best at the left sternal border. Pulses are absent in the distal right extremity. The right femoral pulse and pulses in the left lower extremity are normal. The right lower extremity is pale, cold, and tender with decreased sensation to light touch. Movement of the affected extremity is limited due to pain. Duplex ultrasonography shows thrombi in the left superficial femoral and popliteal veins and an embolus at the right femoral bifurcation. Intravenous heparin therapy is begun. In addition to a right femoral embolectomy, which of the following is the most appropriate next step in management?

- ☐ A) Echocardiography
- ☐ B) Cardiac catheterization
- ☐ C) Placement of an inferior vena cava filter
- ☐ D) Thrombectomy of the left lower extremity
- ☐ E) Tubal ligation



18. A 27-year-old primigravid woman at 10 weeks' gestation comes for her first prenatal visit. She has a history of multiple sexually transmitted diseases and intravenous drug use. Examination shows no abnormalities. HIV testing is recommended, but the patient refuses. Which of the following is the most appropriate course of action for the physician?
- ☐ A) Repeat recommendation of HIV testing in 6 weeks
 - ☐ B) Offer prophylactic zidovudine (AZT) therapy
 - ☐ C) Recommend an elective abortion because she most likely has HIV infection
 - ☐ D) Perform HIV testing without her consent because she is at high risk for HIV
 - ☐ E) Refuse to treat the patient if she does not follow the recommended tests

19. A 19-year-old woman comes to the physician because of a 3-day history of urinary urgency and frequency and pain with urination. She has not had fever, chills, or flank pain. She has no history of serious illness. Menses occur at regular 30-day intervals. Her last menstrual period was 19 days ago. She is sexually active with one male partner and uses an oral contraceptive. Vital signs are within normal limits. Pelvic examination shows mucopurulent discharge on the cervix; there is no vaginal discharge. Bimanual examination shows a normal-sized uterus and no adnexal masses. Microscopic examination of a urine sample shows 1–2 RBCs, numerous WBCs, and no bacteria. Which of the following is the most appropriate pharmacotherapy?
- ☐ A) Acyclovir
 - ☐ B) Doxycycline
 - ☐ C) Fluconazole
 - ☐ D) Metronidazole
 - ☐ E) Zidovudine (AZT)

20. A 65-year-old woman comes to the office because she has had increasing abdominal girth over the past 6 months. Pelvic examination shows a 20-cm mass filling the pelvis; the mass is cystic to palpation. Ultrasonography shows a 23-cm cystic mass with multiloculated cysts that appears to involve the left ovary. Which of the following is the most likely cause of these findings?
- ☐ A) Follicular cyst of the ovary
 - ☐ B) Ovarian tumor of endometrioid origin
 - ☐ C) Ovarian tumor of epithelial origin
 - ☐ D) Ovarian tumor of germ cell origin
 - ☐ E) Ovarian tumor of stromal origin

21. A 37-year-old woman with a 15-year history of endometriosis comes to the physician to request information about undergoing total abdominal hysterectomy and bilateral salpingo-oophorectomy. She continues to have severe pelvic pain despite treatment with various pain medications. Her mother has type 2 diabetes mellitus, and her father has prostate cancer. She read that removal of her uterus and ovaries will help control her pain. She says that she does not expect to have children because she has been unable to conceive for 8 years despite undergoing extensive treatment for infertility. During the next 10 years, this patient will be at greatest risk for which of the following if she undergoes the operation?
- ☐ A) Breast cancer
 - ☐ B) Colon cancer
 - ☐ C) Dementia, Alzheimer type
 - ☐ D) Heart disease
 - ☐ E) Osteoporosis

22. A 37-year-old primigravid woman at 28 weeks' gestation comes to the physician for a routine prenatal visit. Amniocentesis for fetal karyotype at 16 weeks' gestation showed no abnormalities, and she received Rh_o(D) immune globulin intramuscularly. Vital signs are within normal limits. Physical examination shows no abnormalities. Her blood group is B, Rh-negative. The fetal heart rate is 148/min. Which of the following is the most appropriate next step to prevent Rh sensitization in this patient?
- ☐ A) Measurement of cerebral artery flow by Doppler velocimetry
 - ☐ B) Serum screening for atypical antibodies
 - ☐ C) Intramuscular administration of Rh_o(D) immune globulin
 - ☐ D) Amniocentesis for anti-D antibody
 - ☐ E) No further treatment is indicated

23. A 15-year-old girl comes to the emergency department because of the sudden onset of severe pain in the right lower quadrant of the abdomen that began 2 hours ago. Her menses started the day before; her previous menses began 6 weeks ago. Her temperature is 37.5°C (99.5°F). She has right adnexal tenderness. Laboratory tests, including complete blood count, show normal findings. Serum β -hCG testing is negative. Which of the following is the most likely diagnosis?
- ☐ A) Appendicitis
 - ☐ B) Endometrioma
 - ☐ C) Endometriosis
 - ☐ D) Intestinal colic
 - ☐ E) Ruptured ovarian cyst
 - ☐ F) Spontaneous abortion

24. A 27-year-old woman, gravida 1, para 1, comes to the physician for counseling prior to conception. She would like to conceive within the next year, but she and her 38-year-old husband are concerned about the risk for fetal death. Her first pregnancy ended in the delivery of a 3175-g (7-lb) stillborn infant at 34 weeks' gestation. The infant appeared normal, and autopsy did not show any congenital malformations. There was no evidence of an umbilical cord compression or fetal-maternal hemorrhage. She and her husband are both healthy, take no medications, and are in a stable marital relationship. Neither has used tobacco, alcohol, or illicit drugs. Her 37-year-old brother has type 1 diabetes mellitus, and her 60-year-old father has type 2 diabetes mellitus. She is 170 cm (5 ft 7 in) tall and weighs 65 kg (143 lb); BMI is 22 kg/m². Her blood pressure is 102/64 mm Hg. A grade 1/6, soft systolic murmur is heard. Physical examination, including pelvic examination, shows no other abnormalities. Which of the following is the most appropriate advice for this patient prior to conception?
- ☐ A) HLA typing of the patient and her husband
 - ☐ B) Karyotype analysis of the patient and her husband
 - ☐ C) Measurement of the patient's fasting serum glucose concentration
 - ☐ D) Semen analysis
 - ☐ E) Hysterosalpingography



25. A previously healthy 32-year-old woman comes to the physician because of a 4-month history of increasing pain of both breasts. The pain is worse immediately prior to the onset of her menstrual periods. She states that she does not perform monthly breast self-examinations because of the pain and numerous lumps. A paternal aunt has a history of breast cancer. The patient takes no medications. Menses occur at regular 30-day intervals; her last menstrual period was 16 days ago. Examination of the breasts shows bilateral diffuse tenderness and nodularity; no masses can be palpated. There is no galactorrhea, nipple discharge, or axillary adenopathy. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Second examination on day 9 of her next menstrual cycle
 - ☐ B) Measurement of serum prolactin concentration
 - ☐ C) Mammography
 - ☐ D) Ultrasonography of the breasts
 - ☐ E) MRI of the breasts

26. A 3-year-old girl is brought to the physician by her father because of vaginal discharge for 2 days. She has been scratching her vulva and appears uncomfortable. There is a moderate amount of bloody, foul-smelling vaginal discharge. Which of the following is the most likely cause of this condition?

- ☐ A) Adenocarcinoma of the vagina
- ☐ B) Candidiasis
- ☐ C) Foreign body
- ☐ D) Sarcoma botryoides
- ☐ E) Trauma

27. A 38-year-old woman, gravida 3, para 2, at 12 weeks' gestation is admitted to the hospital because of a 1-day history of heavy vaginal bleeding and moderate abdominal pain. She also feels jittery and as if her heart were racing. Her previous pregnancies were uncomplicated and ended in vaginal deliveries at term. Her last menstrual period was 3 months ago. Menses previously occurred at irregular 3- to 4-month intervals. The patient has no history of serious illness and takes no medications. She is sexually active with one male partner; they use condoms inconsistently. Her temperature is 37.8°C (100°F), pulse is 110/min, and blood pressure is 90/50 mm Hg. Abdominal examination shows no abnormalities. The uterus is consistent in size with a 16-week gestation. No fetal heart tones are heard. Pelvic examination shows a large amount of dark blood in the vaginal vault with bleeding from the nondilated cervix. Her hematocrit is 29%. Which of the following is the most likely diagnosis?

- ☐ A) Abruptio placentae
- ☐ B) Ectopic pregnancy
- ☐ C) Endometritis
- ☐ D) Hydatidiform mole
- ☐ E) Placenta accreta
- ☐ F) Uterine rupture
- ☐ G) Vasa previa

28. A 32-year-old primigravid woman at 15 weeks' gestation has had persistent nausea and vomiting for 7 weeks. She has a 6-year history of infertility. This pregnancy is the result of ovulation-induction therapy with clomiphene. The uterine fundus is at the umbilicus. Which of the following is the most likely diagnosis?

- ☐ A) Hydatidiform mole
- ☐ B) Leiomyomata uteri
- ☐ C) Multiple gestation
- ☐ D) Polyhydramnios
- ☐ E) Incorrect dates

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- ☐ B) Leiomyomata uteri
- ☐ C) Multiple gestation
- ☐ D) Polyhydramnios
- ☐ E) Incorrect dates

29. A 32-year-old woman, gravida 3, para 2, at 40 weeks' gestation gives birth to a 3685-g (8-lb 2-oz) newborn by vaginal delivery. Apgar scores are 9 and 9 at 1 and 5 minutes, respectively. Her pregnancy was complicated by a low-lying anterior placenta with no evidence of placenta previa. Her two previous pregnancies ended in cesarean deliveries with low transverse incisions. Fifteen minutes after birth, the placenta has yet to deliver. Gentle firm traction is ineffective, and there is no evidence of bleeding. After 20 minutes, continuous firm traction on the umbilical cord causes a firm fleshy mass to appear through the introitus. In addition to arranging for anesthesia and placing an intravenous catheter, which of the following is the most appropriate next step in management?
- ☐ A) Immediate injection of methylergonovine
 - ☐ B) Immediate replacement of the uterus
 - ☐ C) Curettage of placental remnants from the visible inverted uterus
 - ☐ D) Careful dissection of the placenta from the visible inverted uterus
 - ☐ E) Transvaginal hysterectomy

30. A 27-year-old primigravid woman at 8 weeks' gestation comes to the physician for her first prenatal visit. She has no history of serious illness. Her immunizations are up-to-date except for the measles-mumps-rubella (MMR) vaccine. She received a tetanus vaccine 2 weeks ago because she stepped on a nail. She maintains a vegan diet and eats only organically grown foods. In addition to a prenatal vitamin, she takes daily calcium, zinc, vitamin A, vitamin B₁ (thiamine), beeswax, and rose hip tea. Each day, she walks 3 miles and practices yoga for 1 hour. Examination shows no abnormalities. Which of the following is the most appropriate recommendation regarding prenatal care for this patient?
- ☐ A) Moderation of her exercise routine
 - ☐ B) Targeted ultrasonography because of exposure to tetanus toxoid
 - ☐ C) Decreasing her vitamin A intake
 - ☐ D) Decreasing her vitamin B₁ intake
 - ☐ E) MMR vaccine

31. A 27-year-old primigravid woman with severe asthma has an intrauterine fetal death at 24 weeks' gestation and requests induction. She is given 6.5 L of 5% dextrose in water and a total of 36 units of oxytocin over the following 18 hours. She has the sudden onset of a generalized tonic-clonic seizure. Which of the following is the most likely diagnosis?
- ☐ A) Amniotic fluid embolism
 - ☐ B) Eclampsia
 - ☐ C) Epilepsy
 - ☐ D) Hypocalcemia
 - ☐ E) Hyponatremia

32. A 27-year-old woman, gravida 1, para 1, comes to the emergency department because of a 1-day history of fever and malaise. She says that she is exhausted and has a decreased appetite. She is breast-feeding and has had redness and discomfort of her right breast during this time. Four weeks ago, she underwent forceps-assisted vaginal delivery of a healthy newborn who was recently treated for thrush. She has no history of serious illness and takes no medications. Her temperature is 38.4°C (101.1°F), pulse is 110/min, respirations are 16/min, and blood pressure is 110/65 mm Hg. There is no lymphadenopathy. Examination shows erythema, induration, and tenderness of the outer quadrant of the right breast. Both nipples are cracked and excoriated. The remainder of the examination shows no abnormalities. Which of the following is the strongest predisposing factor for this patient's mastitis?

- ☐ A) Dietary changes
- ☐ B) Incomplete breast emptying
- ☐ C) Newborn thrush
- ☐ D) Nipple excoriation
- ☐ E) Relative immune compromise

33. Three hours after admission to the hospital in labor, a 32-year-old woman, gravida 3, para 1, aborta 1, at 39 weeks' gestation develops fetal bradycardia. The fetal heart rate decreases from a baseline of 150/min to 60/min with no return to baseline after 8 minutes. The pregnancy had been uncomplicated. Amniotomy was performed after admission, and she has been receiving continuous epidural analgesia for pain control. Her first child was delivered by low transverse cesarean section for breech presentation. Her pulse is 108/min, respirations are 14/min, and blood pressure is 100/60 mm Hg. The cervix is 100% effaced and 9 cm dilated; the vertex is at -3 station. Before onset of the bradycardia, the vertex had been at +1 station. Which of the following is the most appropriate next step in management?
- ☐ A) Ephedrine therapy
 - ☐ B) Magnesium sulfate tocolysis
 - ☐ C) Terbutaline therapy
 - ☐ D) Forceps delivery
 - ☐ E) Laparotomy

34. A 32-year-old woman, gravida 3, para 2, at 34 weeks' gestation comes to the emergency department 2 hours after the sudden onset of vaginal bleeding. Her pregnancy had been uncomplicated. Examination shows a uterus consistent in size with a 34-week gestation with normal tone. The fetal heart rate pattern is normal. Contractions are irregular. Over the next hour, she has increasingly severe pain with contractions. Her pulse is 82/min, respirations are 14/min, and blood pressure is 170/110 mm Hg. The fetal heart rate is 150 to 160/min. Deep tendon reflexes are 4+. Laboratory studies show:

Hemoglobin	10.1 g/dL
Leukocyte count	14,000/mm ³ with a normal differential
Platelet count	96,000/mm ³
Serum	
AST	120 U/L
Lactate dehydrogenase	840 U/L
Urine protein	3+

Which of the following is the most likely cause of her vaginal bleeding?

- ☐ A) Abruptio placentae
- ☐ B) Bloody show
- ☐ C) Disseminated intravascular coagulation
- ☐ D) Placenta previa
- ☐ E) Vasa previa

35. A previously healthy 27-year-old woman, gravida 2, para 2, comes to the physician because of a 1-week history of a tender mass at the entrance to the vagina. During this time, she also has had discomfort in this area during sexual intercourse. She is sexually active with one male partner, and they use condoms inconsistently. Pelvic examination shows a 4-cm, fluctuant, moderately tender mass located at the 7-o'clock position relative to the opening of the vagina. Which of the following is the most likely cause of these findings?
- ☐ A) Bacterial infection
 - ☐ B) Fungal infection
 - ☐ C) Neoplasia
 - ☐ D) Trauma
 - ☐ E) Viral infection

36. A 22-year-old woman comes to the emergency department because of a 12-hour history of severe right lower quadrant abdominal pain. Her last menstrual period was 8 weeks ago. Menses had previously occurred at regular 28-day intervals. Abdominal examination shows right lower quadrant tenderness. The uterus is consistent in size with an 8-week gestation. There is right adnexal tenderness. Her quantitative serum β -hCG concentration is 10,000 mIU/mL. Transvaginal ultrasonography shows a gestational sac in the uterus containing a yolk sac; there is a 3-cm, complex right adnexal mass. Which of the following is the most likely diagnosis?
- ☐ A) Corpus luteum cyst
 - ☐ B) Ectopic pregnancy
 - ☐ C) Endometrioma
 - ☐ D) Hydatidiform mole
 - ☐ E) Ovarian germ cell tumor

37. A 42-year-old woman comes to the office because of a 1-year history of progressively irregular and increasingly painful menstrual periods. During this time, she also has had pain during sexual intercourse. Menses occur at 18- to 23-day intervals and last 10 days. Menses previously occurred at regular 28-day intervals. The patient has no history of serious illness and takes no medications. She is sexually active and monogamous with her husband, who had a vasectomy 10 years ago. The patient is 168 cm (5 ft 6 in) tall and weighs 68 kg (150 lb); BMI is 24 kg/m². Her temperature is 37.1°C (98.8°F). Physical examination shows uterine tenderness to palpation. The remainder of the examination shows no abnormalities. An endometrial biopsy specimen obtained 20 days after her last menstrual period shows secretory endometrium with plasma cells. Which of the following is the most likely diagnosis?

- ☐ A) Chronic endometritis
- ☐ B) Endometrial cancer
- ☐ C) Endometrial hyperplasia
- ☐ D) Hypergonadotropic hypogonadism
- ☐ E) Hypogonadotropic hypogonadism
- ☐ F) Pelvic inflammatory disease



Previous



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<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

38. Thirty minutes after undergoing forceps-assisted vaginal delivery of a healthy newborn at term, a hospitalized 19-year-old woman, gravida 1, para 1, reports a 15-minute history of increasingly severe vulvar pain. The infant weighed 3345 g (7 lb 6 oz) at birth. Pregnancy was uncomplicated, and she received routine prenatal care. She has no history of serious illness, and her only medication is a prenatal vitamin. Pelvic examination shows a 5-cm fluctuant, tender, discolored mass on the left labium majus. Examination is consistent with vulvar hematoma. There is no vaginal hematoma or edema. Laceration of which of the following arteries is the most likely cause of this patient's current condition?
- ☐ A) Inferior gluteal
 - ☐ B) Pudendal
 - ☐ C) Superior vesical
 - ☐ D) Uterine
 - ☐ E) Vaginal

39. A 20-year-old woman comes to the office because she has been amenorrheic for 6 months. She has had increasing time intervals between menses over the past 2 years. She has no history of serious illness. She uses a diaphragm for contraception. She is 165 cm (5 ft 5 in) tall and weighs 49 kg (108 lb); BMI is 18 kg/m². Her blood pressure is 100/60 mm Hg. Examination shows normal findings. A urine β -hCG test is negative. Which of the following is the most likely cause of her amenorrhea?
- ☐ A) Abnormal peripheral estrogen conversion
 - ☐ B) Decrease in serum cortisol concentration
 - ☐ C) Increase in serum prolactin concentration
 - ☐ D) Increase in serum thyroid-stimulating hormone concentration
 - ☐ E) Irregular secretion of gonadotropin

40. A 72-year-old woman comes to the physician because of a 2-month history of progressive nausea and vomiting and increasing abdominal girth. Bowel sounds are hyperactive with high-pitched rushes. There is shifting dullness to palpation. Bimanual examination shows a right-sided pelvic mass. Which of the following is the most likely cause of these findings?
- ☐ A) Colon cancer
 - ☐ B) Endometrial cancer
 - ☐ C) Gastric cancer
 - ☐ D) Lymphoma
 - ☐ E) Ovarian cancer

41. A 25-year-old primigravid woman at 28 weeks' gestation has had extreme fatigue, anorexia, and intermittent vomiting over the past 4 weeks. Her pregnancy had been uncomplicated. She appears pale. Fundal height is 28 cm. Her hemoglobin concentration is 8 g/dL, mean corpuscular volume is 105 μm^3 , and mean corpuscular hemoglobin concentration is 25% Hb/cell. Which of the following is the most likely diagnosis?
- ☐ A) Folic acid deficiency
 - ☐ B) Iron deficiency
 - ☐ C) Pernicious anemia
 - ☐ D) Sickle cell disease
 - ☐ E) Thalassemia

42. A 32-year-old woman, gravida 3, para 2, at 40 weeks' gestation is admitted to the hospital in labor. Her pregnancy has been uncomplicated. She has a history of two uncomplicated vaginal deliveries. Her temperature is 37°C (98.6°F), pulse is 80/min, and blood pressure is 90/60 mm Hg. Physical examination shows no abnormalities. The fetal heart rate is reassuring. Labor progresses normally with epidural anesthesia. When the patient is 9 cm dilated and 90% effaced with the vertex at +2 station, she suddenly begins gasping for air and has a generalized tonic-clonic seizure. Examination shows bleeding from the gums, nose, and intravenous sites; there is brisk, bright-red vaginal bleeding. Within 1 minute of beginning the seizure, she has cardiac arrest. Which of the following is the most likely cause of this patient's symptoms?
- ☐ A) Amniotic fluid embolus
 - ☐ B) Eclampsia
 - ☐ C) Hepatic rupture
 - ☐ D) Intravascular release of bacterial endotoxins
 - ☐ E) Spasm in the left anterior descending coronary artery
 - ☐ F) Thromboembolism in the pulmonary artery
 - ☐ G) Thrombosis of cerebral veins

43. A 61-year-old woman comes to the physician because of an itchy rash on her left breast that she first noticed 6 months ago. She says the rash began as a small area of flesh-colored "bumps" that intermittently felt irritated. Since then, the rash has spread and become scaly. She uses powder in her bras because she tends to perspire. She has begun using a different soap. Neither effort has provided relief. She has a 20-year history of hypertension, 5-year history of type 2 diabetes mellitus, and a 59-year history of eczema. Her medications are valsartan and metformin. Vital signs are within normal limits. A photograph of the left breast is shown. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) MRI of the left breast
- ☐ B) Recommendation for use of unscented soap
- ☐ C) Tissue biopsy
- ☐ D) Topical antifungal therapy
- ☐ E) Topical corticosteroid therapy



44. An 80-year-old nursing home resident is brought to the physician because of perineal bleeding for 2 weeks. She reports that she has had something falling out of her vagina for 10 years. A 10 x 6-cm mass with a 1-cm ulcer on the dependent portion is protruding from the area of the vagina. Which of the following is the most likely diagnosis?
- ☐ A) Bartholin duct cyst
 - ☐ B) Complete procidentia
 - ☐ C) Urethral caruncle
 - ☐ D) Vaginal myoma
 - ☐ E) Vulvar carcinoma



45. A 16-year-old girl comes to the office because she has not had a menstrual period in 2 months. Menses previously occurred at regular 30-day intervals. One month ago, she was diagnosed with rubella; the symptoms resolved spontaneously. She has no other history of serious illness and takes no medications. She is sexually active with one male partner; they use condoms inconsistently. Vital signs are within normal limits. Examination shows a uterus consistent in size with an 11-week gestation. A urine pregnancy test is positive. This patient's fetus is at greatest risk for which of the following cardiac disorders?
- ☐ A) Anomalous congenital coronary artery
 - ☐ B) Congenital hypertension
 - ☐ C) Cyanotic heart disease
 - ☐ D) Patent ductus arteriosus
 - ☐ E) Pulmonary artery dilation

46. A 46-year-old woman, gravida 2, para 2, comes to the office because of a 3-month history of painless vaginal bleeding that has required the use of a sanitary pad or tampon. Menarche was at the age of 13 years. Menses occur at 24- to 32-day intervals. She has no history of serious illness and takes no medications. Her mother was diagnosed with breast cancer at the age of 68 years. Vital signs are within normal limits. Pelvic examination shows a mildly enlarged uterus. There is blood from the cervical os. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?
- ☐ A) Combination oral contraceptive therapy
 - ☐ B) Dilatation and curettage
 - ☐ C) Endometrial biopsy
 - ☐ D) Hysteroscopy
 - ☐ E) Progestin-only oral contraceptive therapy
 - ☐ F) Reassurance and follow-up examination in 6 months

47. A 32-year-old woman, gravida 2, para 1, at 10 weeks' gestation comes for her first prenatal visit. She delivered her last child 9 months ago; the pregnancy was complicated by deep venous thrombosis (DVT) of the left lower extremity documented by venous Doppler ultrasonography. Her mother and sister also had DVT during pregnancy, and her maternal grandfather and uncle died of cerebral infarction in their 40s. Examination shows a uterus consistent in size with a 10-week gestation. Which of the following is the most appropriate screening test for this patient?
- ☐ A) Measurement of plasma fibrinogen concentration
 - ☐ B) Measurement of prothrombin and partial thromboplastin times
 - ☐ C) Measurement of serum factor VIII antigen concentration
 - ☐ D) Factor V Leiden mutation analysis
 - ☐ E) Repeat venous Doppler ultrasonography

48. A healthy 27-year-old woman, gravida 2, para 1, at 24 weeks' gestation comes for a routine prenatal visit. Pregnancy has been uncomplicated. She feels well, and fetal movement has been appropriate. Her blood group is A, Rh-positive. Her only medication is multivitamins. At her first prenatal visit, her hemoglobin concentration was 11.2 g/dL. Rubella testing showed immunity, and hepatitis and HIV antibody tests were negative. Urine cultures and testing for *Neisseria gonorrhoeae* and *Chlamydia trachomatis* were negative, and a Pap smear showed no abnormalities. Ultrasonography 1 month ago showed a fetus consistent in size with a 20-week gestation. Examination today shows a uterine fundus measuring 25 cm from symphysis to fundus. The fetal heart rate is 155/min. Urinalysis is within normal limits. Which of the following is the most appropriate recommendation for this patient at this visit?
- ☐ A) Antibody screening
 - ☐ B) 1-Hour glucose tolerance test after a 50-g oral glucose load
 - ☐ C) Repeat testing for *N. gonorrhoeae* and *C. trachomatis*
 - ☐ D) Repeat urine culture
 - ☐ E) Repeat ultrasonography

49. A 15-year-old girl is examined at a juvenile detention facility clinic for an inmate health examination. She has no history of serious illness and takes no medications. Immunizations are up-to-date. She was sexually active with one male partner 3 months ago; they used condoms inconsistently. Vital signs are within normal limits. She is at the 90th percentile for height and 75th percentile for weight. Examination shows no abnormalities. Which of the following is the most appropriate recommendation for this patient?
- ☐ A) Hepatitis B surface antigen and hepatitis B surface antibody assays
 - ☐ B) Human papillomavirus nucleic acid testing
 - ☐ C) Pap smear
 - ☐ D) Rapid plasma reagin
 - ☐ E) Repeat hepatitis B vaccine series
 - ☐ F) Testing for *Chlamydia trachomatis* and *Neisseria gonorrhoeae*

50. A 20-year-old woman, gravida 1, para 1, comes to the physician because of vaginal discharge since an uncomplicated vaginal delivery of her newborn at term 1 week ago. Initially the discharge was red, but it is now white. She is breast-feeding without difficulty. Her temperature is 37°C (98.6°F). Physical examination shows no abnormalities. Pelvic examination shows a thick, white vaginal discharge. No vaginal lacerations are noted. The cervix is closed, and the uterus is consistent in size with a 12-week gestation. Which of the following is the most likely diagnosis?
- ☐ A) Bacterial vaginitis
 - ☐ B) Candidiasis
 - ☐ C) Lochia alba
 - ☐ D) Retained products of conception
 - ☐ E) Undiagnosed vaginal laceration